

**Q1. SET**

ANSWER

```
SUBSET:  
A={1,2}  
B={1,2,3,4}  
print(A.issubset(B))  
OUTPUT:True  
  
SUPERSET:  
A={1,2,3,4}  
B={1,2}  
print(A.issuperset(B))  
OUTPUT:True
```

**Q2. SET**

ANSWER

Union:  
Combines all elements from both sets without duplicates.  
Intersection:  
Gives the common elements present in both sets.  
Difference:  
Gives the element present in one set but not in the other

input  
A: 1 2 3 4  
B: 3 4 5 6  
output:  
union: {1, 2,3,4,5,6}  
intersection:{3,4}  
difference:{1,2}

**Q3. DICTIONARY**

ANSWER

We can use a dictionary with two keys, "Even" and "Odd", and store the number in their respective groups based on whether the number is divisible by 2.

input:  
1 2 3 4 5 6  
output:  
{'Even': [2,4,6], 'Odd': [1,3,5]}

**Q4. DICTIONARY**

ANSWER

A nested dictionary stores each employee's details inside dictionary. We can access the employee's salary using the employee name and the "Salary" key.

input:  
employee name: jotheesh  
age:23  
output:  
salary:30000

**Q5. DICTIONARY**

ANSWER

A dictionary can be used to store each number as a key and its number of occurrence as the value. This help us easily count how many times each number appears in the list.

```
number=[1,2,2,3,3,3,4,4,4,4]  
freq={}  
for n in number:  
    freq[n]=freq.get(n,0)+1  
print(freq)
```

input: 1 2 2 3 3 3 4  
output: {1: 1, 2: 2, 3: 3, 4: 4}